

Loops and control flow ✓

Control flow - order of execution.

→ Control flow means the order in which JS executes statement

Default → Top to Bottom

normally

⇒ main control flow concept ✓

✓ Conditional statement ✓ lecture 3

✓ loops ✓
• break ✓
• continue ✓
• Switch ✓

Loops -

→ A loop allows us to execute the same block of code repeatedly.

Ex { for (let i = 1; i <= 15; i++)
console.log("Satyam")
3

4 Types of Loops

① for Loop ^{keyword}

→ Syntax

for (initialization; conditions; increment/decrement) {
// code
3;
}

Start कहाँ से करना है

रुकना कहाँ है

Steps

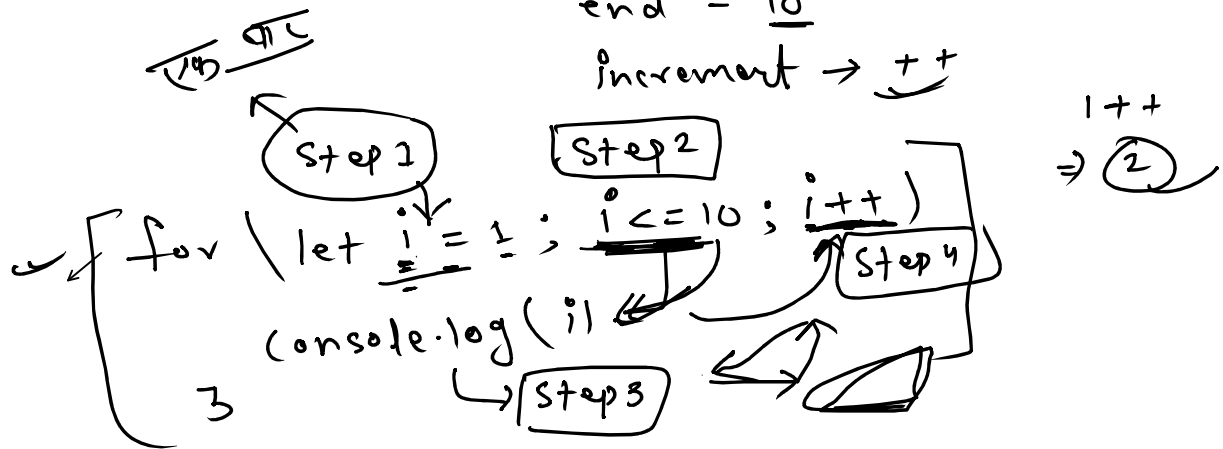
- initialization → Runs once at the beginning
- condition → checked before every iteration
↓
कितनी बार loop चला
- Increment/decrement
↳ Runs after each iteration

⇒ 1 to 10 prints →

Start - 1 ✓

end - 10 ✓

increment → ++



While Loop - A while loop executes as long as the condition is true.

Syntax →

```
while (condition) {  
  // code  
  increment / decrement  
}
```

Starting point →

```
let i = 1;
```

```
while (i <= 10) {
```

```
  console.log(i)
```

```
  i++
```

```
}
```

do while →

→ do-while loop executes the code at least once, then check the conditions.

Syntax -

Start -

do {

// code

} while (condition);

break

↳ Immediately terminates the loop

```
for (let i = 1; i <= 10; i++) {
```

```
  if (i === 5) {
```

```
    break;
```

```
  }
```

```
  console.log(i)
```

```
}
```

→

1
2
3
4

Continue -

Continue -

↳ it skips the current iteration and moves to the next iteration

```
for (let i = 1; i <= 10; i++) {  
  if (i === 5) {  
    continue;  
  }  
  console.log(i) //
```

3

→

1
2
3
4
6
7
8
9
10

Nested loop -

A loop inside another loop is called nested loop

```
[ for (let i = 1; i <= 5; i++) {  
  for (let j = 1; j <= 5; j++) {  
    console.log(i, j) ←
```

- 3 - 3 -

1, 1
1, 2
1, 3
1, 4

1, 5
2, 1
2, 2